Intro to Robotics and Rotation Matrices

EGR 445/545

GVSU

Outline for today

- what is this class about?
- syllabus
- who is Dr. Krauss?
- mugshots
- intro to rotation matrices
- intro to Python
- learning activity 1

What is this class about?

- Learning to think in coordinate systems
- Understanding robot kinematics
  - Forward and inverse kinematics
- Understanding singularities and Jacobians

Key Idea:

The more you know about **robotics theory** and especially **kinematics**, the better equipped you are for a **productive career** as a **robotics engineer**.

Theory Impacts Industrial Practice

Knowing forward and inverse kinematics can greatly expand what you can do with a robot.

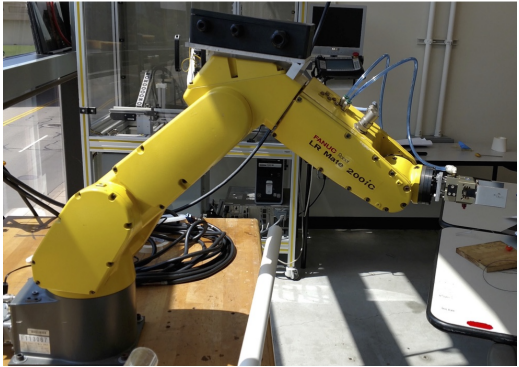
- what if you need to make the robot follow a complex path?
- how will you recognize and avoid singularities?

Driving Questions for Today

- What is the primary challenge in industrial robotics?
- How does robot kinematics help?
- What is a rotation matrix, where do they come from, and how do I use them correctly?

The primary challenge in industrial robotics

- What do I want to know?
- What can I easily measure?



Course Overview

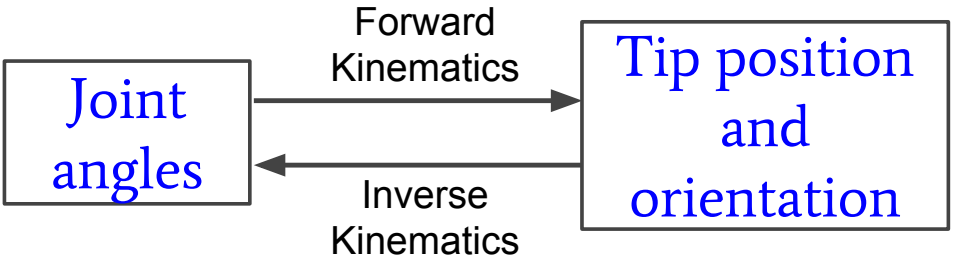
Foundational Message for Chapters 2 and 3

- Having a systematic way of describing the position and orientation of robots with many links will make our lives easier
- there are two major pieces to this systematic approach:
  1. rotation/homogeneous transformation matrices
  2. the Denavit-Hartenberg convention

The Fundamental Challenge in Robotics

- **Challenge**: What makes robotics challenging is that we want to know the position and orientation of the robot in the **global coordinate system**, but what we can measure are the **joint angles**.
- **Want**: global tip position and orientation
- **Have**: joint angles and link lengths

Forward and Inverse Kinematics



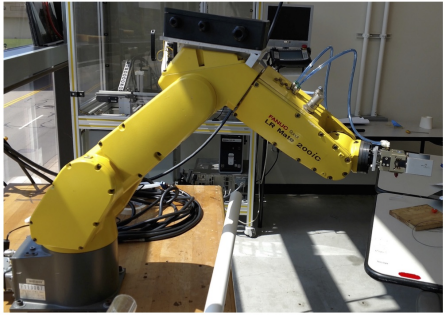
Homogeneous Transformation Matrices

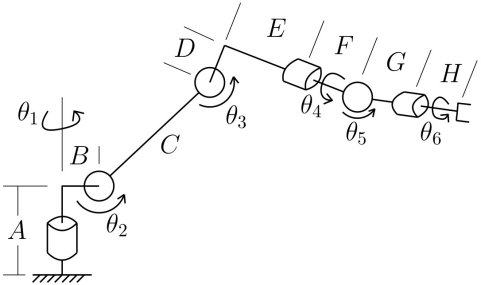
- HT matrices are the building blocks of robot kinematics
- HT matrices allow us to move from one coordinate system to another
  - combining rotation and translation in one matrix multiplication
- attaching a coordinate system to each joint/link of a robot allows us to move from tip to base (global) coordinates

Big Picture Learning Objectives

- I want you to know how to model and analyze robots by attaching coordinate frames to each link. (chap. 2 and 3)
- I want you to know how to move between those frames.
- I want you to use those frames to solve forward and inverse kinematics problems. (chap. 3 and 4)
- I want you to know how to find and use Jacobians to analyze robot velocities and find singularities. (chap. 5)

HT Matrices and Industrial Robots





- if you were give $\theta_1, \theta_2, \ldots\ \theta_6$ and A through H, HT matrices would allow you to find the tip position and orientation
- this would be a very difficult problem to solve using just trig
  - we need the DH convention and HT matrices

Intro to Rotation Matrices

Introduction to rotation matrices

- What is a rotation matrix?
- What do I expect you to learn about them?
- Why can't we just Google "rotation matrix" when we need the formula?

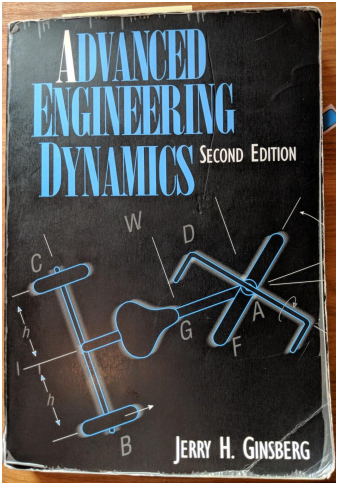
Deriving Rotation Matrices

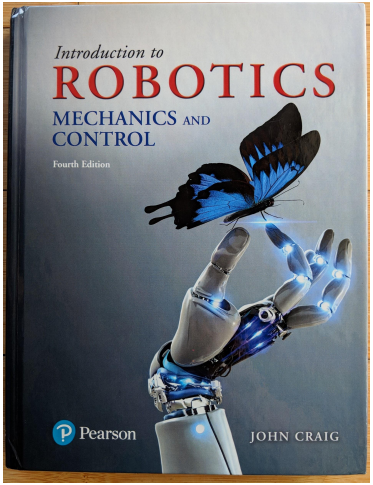
- one picture/sketch
- less than one minute
- knowing how to derive rotation matrices will deepen your understanding
- gives you tools to check your answers
    - especially the signs

Why can't we just Google "rotation matrix"?

- if I need to use a rotation matrix in the future, I will just "google" it
  - what could go wrong?
- you have a 50/50 shot of sign errors

Two Opposing Sign Conventions





Two Opposing Sign Conventions (cont.)

Ginsberg:

$$[R_x] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & \sin\theta_x \\ 0 & -\sin\theta_x & \cos\theta_x \end{bmatrix}$$

$$[R_y] = \begin{bmatrix} \cos\theta_y & 0 & -\sin\theta_y \\ 0 & 1 & 0 \\ \sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$$

$$[R_z] = \begin{bmatrix} \cos\theta_z & \sin\theta_z & 0 \\ -\sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Craig:

$$R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_Y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

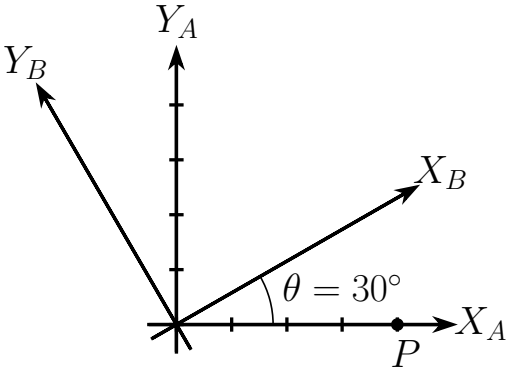
Signs and Rotations Matrices

What happens if we naively "google" rotation matrix and apply it to this problem?

Assume we want to use

$$P_{rotated} = \mathbf{R} \cdot P_{original}$$

- Let's "code this up" and use it as an intro to Python
  - there is a starter notebook in the class 1 github folder



${}^{A}_{B}\mathbf{R}$ vs. ${}^{B}_{A}\mathbf{R}$

- what does ${}^{A}_{B}\mathbf{R}$ do?
- how do we create a sketch for ${}^{A}_{B}\mathbf{R}$?

${}^{A}_{B}\mathbf{R}$ vs. ${}^{B}_{A}\mathbf{R}$ (part 2)

- what does ${}^{B}_{A}\mathbf{R}$ do?
  - how is it different from ${}^{A}_{B}\mathbf{R}$?
- how do we create a sketch for ${}^{B}_{A}\mathbf{R}$?

Usage Conventions

$$^{A}P = \mathbf{R}\,^{B}P$$

or

$$^{B}P = \mathbf{R}\,^{A}P$$

Sign Convention Issues

- Base-to-tip or tip-to-base?
- Space-fixed or body-fixed rotations?
- Anyone who says there is only one definition for a rotation matrix is probably not as smart as they think they are.
- When you memorize or google you are also picking a convention
    - probably without realizing it

Why we are going to learn to derive rotation matrices?:

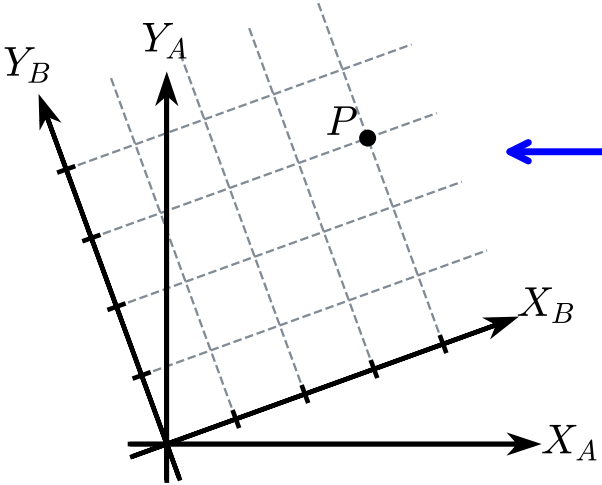
- It isn't hard
- Promotes better understanding/deeper learning
- Protects against sign errors
- You know which convention you are using.

Deeper Understanding

If you understand this, you understand where rotation matrices come from and what they do:

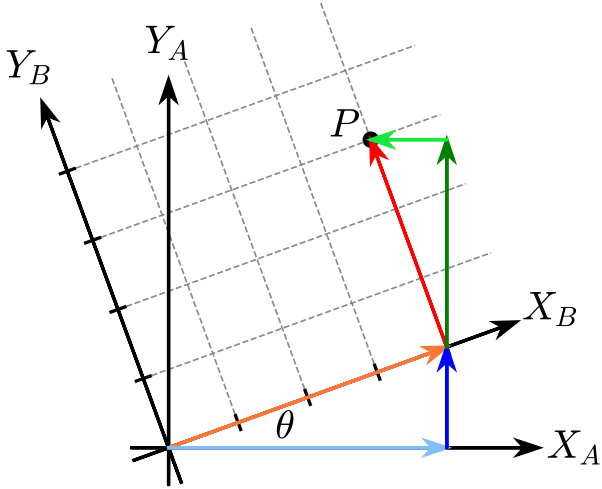
$$
{}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B & {}^{A}\hat{Z}_B \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}
$$

70% of Chapter 2 in one graphic



$$^{A}P = {}^{A}_{B}R\,{}^{B}P$$

Key Sketch for $\mathbf{R}_z$



Learning Activity 1

Learning Activity 1

- download from the student github page